Lead leg ground reaction forces and knee kinematics in the disc golf backhand throw

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ABSTRACT

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The purpose of this study was to describe the characteristics of lead leg ground reaction forces and knee kinematics in the disc golf backhand throw. The popularity of disc golf has been increasing, while research on its biomechanics remains limited.

Ten male disc golfers (age 23,2 ± 2,5) with competition experience ranging from hobby level to international elite level tournaments participated in this study. Each participant performed at least five maximal backhand throws. The trial with the highest launch speed was considered in the analysis. Three-dimensional ground reaction forces of the lead leg were measured on a force plate. The peak vertical, anteroposterior, mediolateral, horizontal resultant and total resultant forces were analyzed. Lead knee movement patterns and knee extension velocity were assessed with a goniometer.

The mean launch speed was 30,7 ± 2,7 m/s. Association with launch speed was observed in five peak force parameters. They were the absolute peak braking force ($r = -0{,}748$, $p = 0{,}013$), the relative peak braking force ($r = -0{,}735$, $p = 0{,}015$), the absolute peak horizontal resultant force ($r = 0{,}732$, $p = 0{,}016$), the relative peak horizontal resultant force ($r = 0{,}734$, $p = 0{,}016$) and the absolute peak total resultant force ($r = 0{,}669$, $p = 0{,}034$). Peak lead knee extension velocity was associated with higher launch speed ($r = 0{,}781$, $p = 0{,}008$) when the follow-through phase was included in the analysis. Three knee movement patterns were identified, but association with launch speed was not analyzed.

Being able to create greater ground reaction forces, particularly in the posterior direction, may help generate a higher launch speed. Greater knee extension velocity was associated with higher launch speed. However, its contribution remains a question, as the timing of peak values varied significantly. The characteristics of ground reaction forces and knee kinematics were presented along with suggestions for future studies.

Keywords: disc golf, launch speed, ground reaction forces, knee kinematics

ABBREVIATIONS

BW	body weight
Frh	horizontal resultant ground reaction force
Frt	total resultant ground reaction force
Fx	mediolateral ground reaction force
Fy	anteroposterior ground reaction force
Fz	vertical ground reaction force
GRF	ground reaction force
PD	proximal-to-distal
PDGA	Professional Disc Golf Association

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1 INTRODUCTION

The base idea of disc golf is very similar to ball golf. The goal is to complete a course in the fewest throws possible using a disc. A course consists of multiple holes (typically nine or eighteen) that begin at a teeing area and end at a basket target. (PDGA 2024b) The two main throwing styles are forehand and backhand, of which the latter is the most commonly used style (Nelson et al. 2015).

According to a survey conducted in 2022, disc golf was the third most popular summer free time sport activity among 11–15-year-old boys in Finland (Martin et al. 2023, 18). However, disc golf is not just a free time activity, as there were over 130 000 active members worldwide of the Professional Disc Golf Association at the end of 2023, while the total number of memberships issued over time was over 250 000. The number of members experienced a rapid increase during Covid, and even though the growth has slowed down, the trend is still upward. (PDGA 2024a)

To the author's knowledge, there are only a handful of studies on the mechanics of disc golf. To name a few, Leppilahti (2019) has conducted a kinematic analysis of the backhand throw and Holcomb (2023) has studied the EMG activation of the upper extremity during a stand still backhand throw. Both studies mentioned were student theses. Most recently, Koch et al. (2024) have investigated the optimal thumb position on the disc. A few other scientific studies have investigated the epidemiology of injuries in disc golf (Nelson et al. 2015; Rahbek & Nielsen 2016).

The purpose of this thesis is to increase scientific understanding of disc golf. The goal is to study the lead leg ground reaction forces and knee kinematics in the backhand throw. As research on disc golf is very limited, the results of this study will be compared to the mechanics of other throwing motions to find the similarities and the differences. This information can be used to understand the physical requirements of disc golf better.

2 BIOMECHANICS OF THE BACKHAND THROW

The disc golf backhand throw has been divided into four major phases by Leppilahti (2019). The movement begins with the preparation phase which includes a run-up. The second to last step of the run-up is a cross step which helps rotate the hip and the thoracic spine away from the target. This is followed by the loading phase, during which the thoracic spine reaches its maximum rotation away from the target. The action phase begins with the lead foot contacting the ground. The foot is planted perpendicular to the target. Most segments reach their maximum angular velocities during the action phase, which ends at disc release. The follow-through phase begins as the disc is released. During this phase, rotational motion begins to decelerate, except for the shoulder and wrist, which reach their maximum angular velocities after the disc is released. The lead leg is also rotated on the heel toward the target to allow proper rotation of the body. (Leppilahti 2019)

This chapter will discuss the biomechanics of the backhand throw from the point of view of the kinetic chain. In addition, the factors associated with the launch speed of the disc are presented. The concept of kinetic chain originates from engineering, but it has been applied to human movement (Karandigar & Vargas 2011). The kinetic chain principle suggests that the body consists of links that are connected to each other and the motion of one link affects the adjacent links. In practical terms, these links are body segments such as feet, trunk and shoulder. (Ellenbecker & Aoki 2020) The kinetic chain can be considered open or closed. In the open chain, the most distal segment can move freely, whereas in the closed chain, the distal segment meets external restriction, or in other words, is fixed. (Karandigar & Vargas 2011). A one-handed tennis backhand stroke is an example of an open kinetic chain movement (Stepień 2011) and so can the disc golf backhand throw be defined as an open movement.

2.1 Proximal-to-distal sequence

The proximal-to-distal (PD) sequence, or order, is used to describe the mechanics of human movement, and it is similar to the kinetic chain principle (Ellenbecker & Aoki 2020). The main idea of the PD order is that, to maximize speed at the endpoint of the sequence, motion should progress from the more proximal segments to the more distal segments. Ideally, the more distal one of two adjacent segments begins its motion as the more proximal segment reaches its

greatest speed. Furthermore, the maximal speed of a segment should exceed the speed of the preceding segment. Thus, the speed should be greatest at the most distal segment of the sequence. These descriptions are based on the *Summation of speed principle* created by John W. Bunn in 1972. (Putnam 1993)

According to Putnam (1993), the PD sequence can be described using the linear velocities of segment endpoints, joint angular velocity or segment angular velocity. Leppilahti (2019) has studied the joint and segment angular velocities in the disc golf backhand throw. Based on the results, it can be stated that the backhand throw follows the principles of the PD order except for the external rotation of the shoulder. The external rotation of the shoulder appears to reach its maximum angular velocity after the elbow and the wrist have reached their maximum velocities. (Leppilahti 2019)

The occurrence of the PD sequence has been studied in multiple sports. It can be clearly recognized in tennis during the serve and the backhand groundstroke (Marshall & Elliott 2000; Stepień et al. 2011). The sequence is also evident in overarm throwing motions such as baseball pitching (Hirashima et al. 2008; Kibler et al. 2013). Contrarily, according to Hummel (2003), the ultimate frisbee backhand throw has been shown to follow the PD sequence only partially as the actions of elbow differ from the theory. However, this result is based on an analysis of only one subject who performed submaximal throws.

The traditional proximal-to-distal sequence theories have faced criticism. Marshall and Elliott (2000) have suggested that the traditional descriptions are missing the long-axis movement analysis such as shoulder internal rotation. In their study, the upper limb followed the PD order in tennis serve and squash forehand drive when the analyzed segmental movements were abduction-adduction and flexion-extension. However, the peak angular velocity of the shoulder internal rotation occurred simultaneously with wrist flexion just prior to impact. (Marshall & Elliott 2000) This sequence can be noticed in baseball pitching as well (Hirashima et al. 2008). According to Putnam (1993), the maximal shoulder angular velocity should occur prior to elbow and wrist movements, although it was mentioned that the sequence is not clear in some movements due to the exact reasons discussed above. In this light, although the disc golf backhand throw differs slightly from the traditional descriptions of the PD order (Leppilahti 2019), the mechanics are similar to other throwing and striking motions.

The kinetic chain and proximal-to-distal sequence are complicated phenomena that would require more explanation. However, the main function of this chapter is to prove that the mechanics of the backhand throw are comparable to overarm throwing and striking motions, although the technique differs significantly. Further discussion is not necessary considering the context of this study.

2.2 Kinematic variables associated with the launch speed of the disc

In disc golf, producing a high launch speed is important when the goal is to throw far, as launch speed is one of the most important variables to determine the final range of the throw, along with launch angles (Kamaruddin 2011, 133). Leppilahti (2019) is so far the only one who has studied the association between launch speed and different kinematic variables in disc golf. In the study, variables that were correlated with launch speed were the maximum angular velocities of shoulder external rotation and thoracic spine rotation as well as the range of motion of thoracic spine and pelvis rotation. (Leppilahti 2019)

In addition, Leppilahti (2019) divided the participants (n = 8) into two groups based on launch speed. The participant with the highest launch speed was used as a reference, to which the results of the other participants (n = 7) were compared to. Statistically significant differences between the groups were launch speed, stance deviation, spin rate of the disc, as well as the maximum angular velocities of spine rotation, thoracic spine rotation and shoulder external rotation. (Leppilahti 2019)

3 GROUND REACTION FORCE

Ground reaction forces have not been studied in disc golf, so this chapter will discuss the characteristics of reaction forces with the help of findings from various movements. Later, the emphasis is put on the effect of peak ground reaction forces, as well as their timing, on throwing performance.

3.1 Overview of ground reaction forces

The ground reaction force (GRF) represents the reaction of the ground to the acceleration of body segments. It is based on Newton's law of action-reaction. (Enoka 2015, 50) The three components of the GRF can be measured with a force plate. The components are vertical, anteroposterior, or the direction of the movement, and mediolateral, or perpendicular to the direction of the movement. (Nigg 2007, 326) The magnitudes of these three components also define the magnitude and direction of the resultant force vector (Enoka 2015, 52), which is illustrated in figure 1.

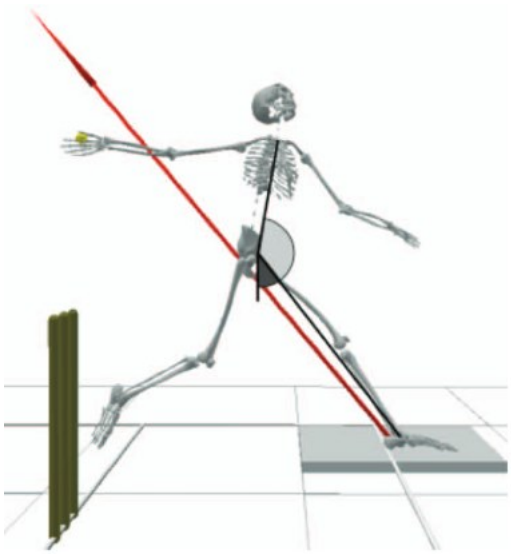

FIGURE 1. An example of the resultant force vector (red) alignment during a cricket bowl. (modified from Worthington et al. 2013b)

The three components are used to describe changes in momentum of the center of mass (Nigg 2007, 327). When the vertical component is greater than body weight (BW), the center of mass

accelerates upwards. On the contrary, the center of mass experiences a downward acceleration when the vertical component is less than body weight. (Enoka 2015, 50) The anteroposterior component can be referred to as propulsive and braking forces. In walking and running, a braking reaction force occurs as the foot contacts the ground in front of the center of mass. Oppositely, a propulsive reaction force occurs when the center of mass is in front of the stance foot and force is applied to the ground to generate forward motion. (Enoka 2015, 50–51; Nigg 2007, 332) The mediolateral component functions in the side-to-side direction, but during walking and running, it is often lesser in magnitude and has more variability compared to the other components (Enoka 2015, 51; Nigg 2007, 332).

In heel-toe running, the vertical component tends to have two peaks, called the impact and active peaks. Generally, in human locomotion, impact forces reach their maximum during the first 50 milliseconds after the contact of two objects. In running, the two objects are a foot and the ground, and the peak can occur as early as 5 milliseconds after the contact. On the other hand, the active forces are generated by movement that is fully controlled by muscular activity. The impact forces can be much greater than the active forces, even though the active forces are greater during some movements, like walking, for example. Impact and active force peaks can occur in the horizontal components as well. (Nigg 2007, 329–331) In disc golf, as the lead foot is planted, the toes contact the ground before the heel (Leppilahti 2019). In forefoot running, toe contact induces a smaller impact peak compared to heel strike (Nigg 2007, 329) or no impact peak at all (Lieberman et al. 2010).

The magnitudes of the GRFs increase as running speed increases (Hamill et al. 1983; Jiang et al. 2024). At running speeds between 10 to 25 km/h, the peak vertical impact force has been shown to range from 1,5 to 3,3 times BW, while the active force ranges from 2,5 to 3 times BW. The active force remains more constant with increasing speeds, which can be seen in figure 2. The anteroposterior forces also increase with running speed. The peak braking forces range from approximately -0,3 to -0,5 times BW. (Hamill et al. 1983; Jiang et al. 2024)

In contrast to the relatively low GRF values found in running, Perttunen et al. (2000) have measured forces over 15 times BW in the vertical direction and 7 times BW in the posterior direction during the triple jump. The peak forces were impact forces, while the active forces were lower in magnitude. The impact force peaks were significantly greater than in running, for example, which can be seen in figure 3.

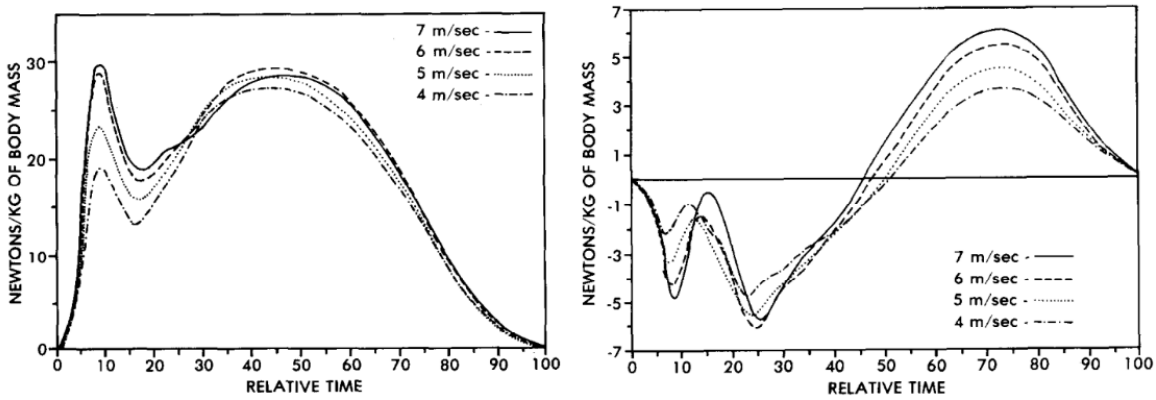

FIGURE 2. Vertical (left) and anteroposterior (right) GRFs during different running speeds. The X-axis presents the time from foot contact to toe-off. (modified from Hamill et al. 1983)

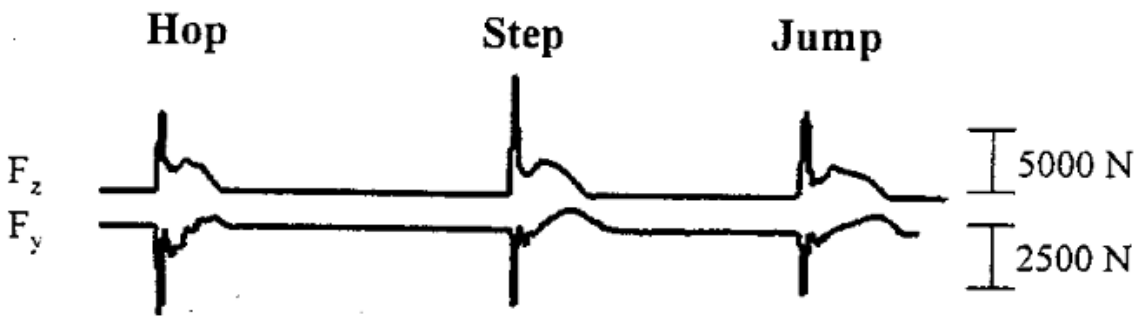

FIGURE 3. Example of two-dimensional ground reaction forces during the triple jump. Fz represents the vertical component while Fy is the anteroposterior component. (Perttunen et al. 2000)

3.2 Ground reaction forces during throwing motions

Moving on to throwing motions, Hurrion et al. (2002) have studied GRFs during the delivery stride of javelin throwing among five male participants. Three successful throws from each participant were analyzed. The peak vertical force was 5,63 times BW and the peak resultant force was 7,05 times BW, on average. On the other hand, Reinholm (2024) has studied elite male and female javelin throwers. On average, the peak vertical force for males was 7,67 times BW and for females 6,72 times BW. The mean peak braking forces were -5,69 times BW for males and -5,10 times BW for females. (Reinholm 2024) The peak resultant forces were not reported by Reinholm (2024), but the peak vertical forces were greater in both males and

females when compared to the results of Hurrion et al. (2002). A similar movement to javelin throwing is cricket bowling considering the kinematics and run-up speed, for example (Worthington et al. 2013b; Reinholm 2024). Portus et al. (2004) have measured GRFs during cricket bowling among 42 elite cricket bowlers. The mean peak braking force was -4,5 times BW and the peak vertical force was 7,3 times BW. Worthington et al. (2013b) has conducted a similar study involving twenty elite cricket bowlers. The mean peak braking force was -4,5 times BW and the peak vertical force was 6,7 times BW. These results were almost identical to the ones reported by Portus et al. (2004).

In baseball, there are a few studies that have investigated the magnitude of stride leg ground reaction forces. The stride leg in pitching is the front leg and it functions as a support for the entire body weight after contacting the ground (Elliott et al. 1988; McNally et al. 2015). McNally et al. (2015) studied GRFs among seventeen adult recreational baseball pitchers. The mean peak braking force was -0,65 times BW while the vertical force was 1,46 times BW. The mean peak resultant force was 1,59 times BW.

In contrast to McNally et al. (2015), greater forces have been reported by other authors. In a study conducted by Kageyama et al. (2014), the participants were assigned to a high-velocity group and a low-velocity group based on their pitching velocity. For the high-velocity group, the mean peak vertical force was 20,6 N/kg and the peak braking force was -11,7 N/kg. These results are equal to 2,1 and -1,2 times BW, respectively. The mean peak resultant force was 23,7 N/kg, which is equal to 2,4 times BW. (Kageyama et al. 2014) Wasserberg and Giordano (2023) have analyzed the ground reaction forces of 105 high-velocity baseball pitchers. The mean peak forces were 1,76 times BW in the posterior direction and 2,93 times BW in the vertical direction. The mean peak resultant force was 3,35 times BW. These results are significantly greater than in the studies mentioned previously, which Wasserberg and Giordano (2023) suggested to be potentially due to methodological differences. However, the pitching velocity was also generally higher in their study compared to many of the prior studies.

A summary of the findings discussed in this chapter is gathered in figure 4. The illustration is indicative as the data is picked from the studies that reported the greatest values in each movement.

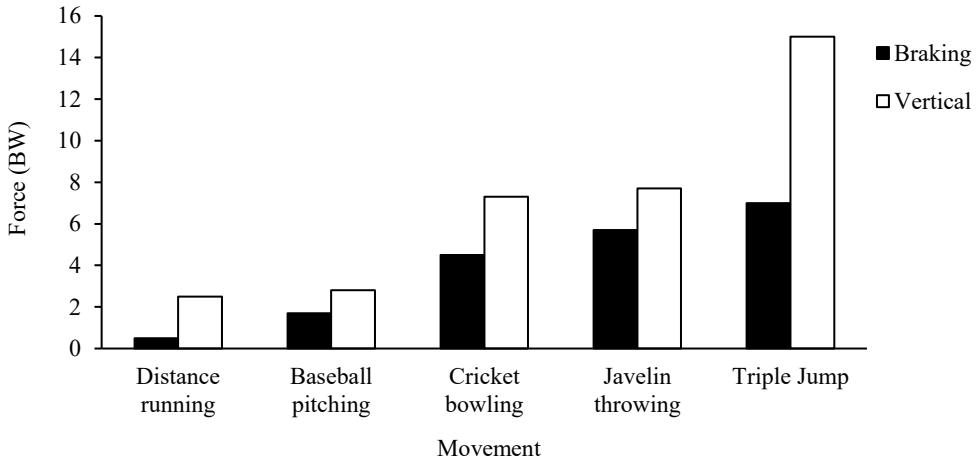

FIGURE 4. Illustration of the body weight (BW) relative peak vertical and peak braking forces during different movements. Braking force is expressed as positive values for clarity. The data is retrieved from Jiang et al. (2024), Wasserberg and Giordano (2023), Portus et al. (2004), Reinholm (2024) and Perttunen et al. (2000).

3.3 Ground reaction forces and throwing performance

In Javelin throwing, research on the association between GRFs and throwing performance is limited. However, Reinholm (2024) found a strong correlation between peak relative braking force and the launch speed as well as the theoretical javelin throwing distance in males but not in females. The peak vertical force did not have a similar correlation in either group. (Reinholm 2024) Among elite cricket bowlers in the study by Portus et al. (2004), the peak braking force was moderately correlated with ball speed. No relationship between peak vertical force and throwing performance was found.

In baseball pitching, studies investigating the association between GRFs and ball velocity have shown varying results. Kageyama et al. (2014) reported significantly greater peak values for every reaction force component in the high-velocity group compared to the low-velocity group. However, the correlations between ball velocity and GRFs were not analyzed. In the study by McNally et al. (2015), the peak vertical, horizontal and resultant forces were strongly correlated with linear wrist velocity during the arm-cocking and arm-acceleration phases. Although the ball velocity was not measured, linear wrist velocity and ball velocity have been shown to have

a high correlation (r^2 = 0,97) (MacWilliams et al. 1998, according to McNally et al. 2015). Contrary to the previously mentioned studies, Wasserberg and Giordano (2023) analyzed the GRFs throughout the pitching cycle instead of collapsing the data to singular peak data points, as they described it. The statistical analysis revealed an association between ball velocity and the braking ground reaction force between 27 and 35 % of the time from foot contact to ball release. No other relationships between the GRF parameters and ball velocity were reported. (Wasserberg & Giordano 2023)

In cricket bowling, Portus et al. (2004) found a moderate negative correlation between ball speed and both time to peak vertical force and time to peak braking force. The average time to peak vertical and peak braking forces ranged from 0,04 to 0,08 seconds across the four different groups of participants. Worthington et al. (2013b) reported a lower average of 0,03 seconds for both variables but relationship with ball speed was not analyzed. In the study by Reinholm (2024), the male and female javelin throwers reached peak forces at approximately 0,03 seconds after foot contact.

Instead of investigating the time to peak force, Kageyama et al. (2014) have studied the relative timing of peak forces in baseball pitching. They found a statistically significant difference between the high-velocity group and the low-velocity group considering the timing of peak vertical and peak resultant force. A later occurrence of these peak forces characterized the high-velocity group compared to the low-velocity group. A similar difference was not observed in the timing of peak braking force. The peak forces occurred at approximately 70 to 85 % of the foot contact phase (Kageyama et al. 2014) Similarly, McNally et al. (2015) observed that the peak vertical and braking forces occur near maximal shoulder external rotation, which is timing-wise in line with the results of Kageyama et al. (2014). These results suggest a significant difference considering the characteristics of ground reaction forces in baseball pitching compared to cricket bowling and javelin throwing.

4 LEAD LEG KNEE KINEMATICS DURING THROWING MOTION

Unlike ground reaction forces, the kinematics of the lead leg have received attention in disc golf research, although only once. In this particular case, Leppilahti (2019) analyzed the knee angle at the time of disc release. The knee angle at disc release was 144 degrees on average, ranging from 112 to 162 degrees. The knee angle was found to be positively correlated with the angular velocity and the range of motion of pelvis rotation. On the contrary, the knee angle had a negative correlation with the range of motion of spine rotation. (Leppilahti 2019) Therefore, the results are slightly conflicting and only little is known about the kinematics of the lead knee in the backhand throw.

4.1 Knee extension velocity

Knee extension velocity has been studied by multiple authors in baseball pitching. In the study by Kageyama et al. (2014), the stride leg knee extension velocity at ball release was significantly higher in the high-velocity group, in which the knee was extending at a velocity of 204,8 ± 100,0 degrees per second (°/s). In the low-velocity group the velocity was 87,1 ± 101,0 °/s. (Kageyama et al. 2014) This result is in line with Chu et al. (2009) who compared male and female pitchers. A significantly greater extension velocity in the male group was observed compared to the female group, while the male pitchers also had greater ball speed. The angular velocity in the male group was 320 ± 180 °/s, which exceeds the results of Kageyama et al. (2014). The angular velocity in the female group was 80 ± 190 °/s (Chu et al. 2009), which is close to the results of the low-velocity group in the study by Kageyama et al. (2014). Matsuo et al. (2001) have also studied knee extension velocity among high-level pitchers, assigning the participants either into a high-velocity or a low-velocity group based on their pitching velocity. The results fall in between the studies above as the extension velocity at ball release was 243 ± 149 °/s in the high-velocity group and 124 ± 141 °/s in the low-velocity group. The difference was statistically significant. (Matsuo et al. 2001)

In addition to extension velocity at release, Kageyama et al. (2014) have analyzed the peak extension angular velocity during the stride foot contact phase. The maximum extension angular velocity was 267,2 ± 98,6 °/s in the high-velocity group and 163,6 ± 129,2 °/s in the low-velocity group. The difference was statistically significant. (Kageyama et al. 2014) These

results were reported in a conflicting way, and it is not definite whether these values were observed during knee extension or flexion. However, based on the conclusions of Kageyama et al. (2014), it is likely that the values did in fact represent greater angular velocity of knee extension. Dowling et al. (2022) have also analyzed knee extension velocity among professional baseball pitchers. The peak extension velocity was statistically significantly greater in the high-velocity group (411 ± 133 °/s) than in the low-velocity group (293 ± 119 °/s). Additionally, peak extension velocity was a predictor of ball speed, although a weak one. (Dowling et al. 2022) The values may not be exact as they were reported slightly differently in the table and the text. According to Chu et al. (2009), a flexing knee throughout the pitch and low extension velocity are indications of not being able to stop the forward movement of the lower body, which ultimately leads to insufficient energy transfer to the trunk.

Kageyama et al. (2014) and Chu et al. (2009) have investigated the timing of peak extension velocity in baseball pitching. Kageyama et al. (2014) found that peak extension velocity occurred later, although not statistically significantly, in the high-velocity group than in the low-velocity group (71,3 % and 55,8 %, respectively). In the study by Chu et al. (2009), the peak extension velocity occurred at 98 % in the male group and at 120 % in the female group. The timing varied significantly between the studies and therefore, making a conclusion is difficult.

4.2 Knee movement patterns

Different knee movement patterns have been identified in overhead throwing motions. In the study by Whiting et al. (1991), 85 javelin throws were analyzed of which the ten longest and ten shortest throws were assigned to groups. Knee movement patterns varied between the longest group and the shortest group. Examples of these patterns are presented in figure 5. Pattern A was typical in the longest group while pattern B was observed in the shortest group. In both groups, knee flexion followed the foot contact, and the first trough (Min1 in figure 5) was almost identical, even though they differ in the example. However, the second trough (Min2) was significantly lower in pattern B, meaning that the shortest group had more knee flexion than the longest group. The amount of knee extension prior to release was almost identical between the groups, but because of greater net flexion during the entire foot contact

phase in the shortest group, the knee was more flexed at release than in the longest group. (Whiting et al. 1991)

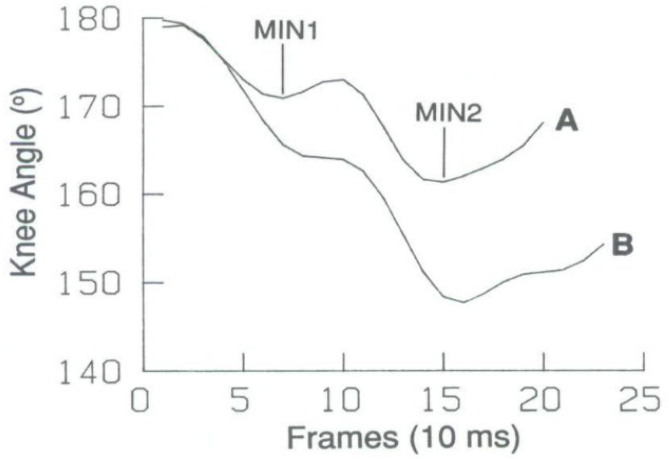

FIGURE 5. Typical knee movement patterns during javelin throwing. Pattern A illustrates the pattern observed in the group with the longest throws and pattern B illustrates the group with the shortest throws. The X-axis presents the time from foot contact to javelin release. (Whiting et al. 1991)

Matsuo et al. (2001) have identified a similar double flexion-extension pattern in baseball pitching as Whiting et al. (2001) have identified in javelin throwing. Three other types of lead knee movement patterns were also identified, with distinct differences between the high-velocity and the low-velocity group. The four different patterns are illustrated in figure 6. The double flexion-extension pattern (type A in figure 6) includes slight knee flexion and extension during the first half of foot contact, and significant extension during the latter half of the throw. This pattern occurred in 69 % of the high-velocity group and in 9 % of the low-velocity group. Type B pattern includes knee flexion during the first half of the throw and slight extension during the last 40 % of the foot contact phase. This pattern occurred in 14 % of the high-velocity group and in 26 % of the low-velocity group. Type C presented alternating knee flexion and extension, ending with knee flexion through ball release. Type D presented a continuous knee flexion pattern. Type C and D were typical in the low-velocity group (26 % and 17 %, respectively) and only C was apparent in the high-velocity group (3 %). (Matsuo et al. 2001)

Knee movement patterns have also been studied in cricket bowling. Portus et al. (2004) classified high-level bowlers into four groups based on the movement patterns. These groups were called flexor, extender, flexor-extender and constant brace. Flexors had at least ten degrees

of flexion followed by less than ten degrees of extension, whereas extenders were the exact opposite. Flexor-extenders had at least ten degrees of both flexion and extension. Constant bracers had less than ten degrees of both flexion and extension. Knee extenders and flexor-extenders generally produced greater ball speeds, but the difference was not statistically significant. However, the net knee extension during the front foot contact phase had a moderate positive correlation with ball speed. (Portus et al. 2004)

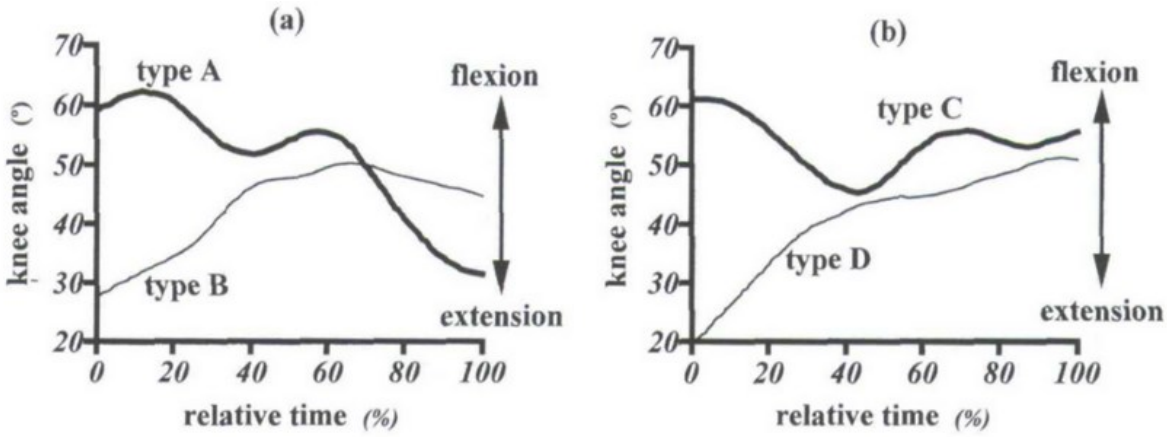

FIGURE 6. Lead knee movement patterns during a baseball pitch. The X-axis presents the time from foot contact to ball release. (Matsuo et al. 2001)

As was mentioned before, Leppilahti (2019) did not analyze the knee angle other than at disc release. However, the knee angle of the participant whose data was used as a reference in the study was further discussed. At foot contact, the knee flexion angle was 47 degrees and at disc release it was 37 degrees. During the action phase, the largest flexion angle was 53 degrees. (Leppilahti 2019) Based on this information it is impossible to tell if the movement pattern was identical to the double flexion-extension pattern presented by Whiting et al. (1991) and Matsuo et al. (2001), but at least there appears to be some similarity. It is worth noting that in the backhand throw, the front foot tends to point perpendicular to the target after ground contact and rotates towards the target after disc release (Leppilahti 2019). Whereas in baseball, for example, the foot is pointing forward throughout the throw (Kibler et al. 2013). This is a considerable difference when comparing the knee kinematics in an overhead throw and a disc golf backhand throw.

5 RESEARCH QUESTIONS

The main objective of this study was to analyze the characteristics of lead leg ground reaction forces and knee kinematics during the disc golf backhand throw. Two research questions and hypotheses were developed to explore the relationship between the peak ground reaction forces and knee extension velocity, with launch speed. The timing of peak forces and peak extension velocity were additionally investigated. The main objective considering the knee kinematics was to describe the lead knee movement patterns during the action phase.

Research question 1: Are the peak ground reaction forces associated with launch speed?

Hypothesis: Yes. In baseball pitching, greater braking, vertical and resultant forces have been shown to be associated with better throwing performance (Kageyama et al. 2014; McNally et al. 2015). In cricket bowling and javelin throwing, greater braking forces have been shown to be associated with better throwing performance (Portus et al. 2004; Reinholm et al. 2024).

Research question 2: Is the knee extension velocity associated with launch speed?

Hypothesis: Yes. Greater knee extension velocity at ball release has been shown to be associated with greater ball speed in baseball pitching (Chu et al. 2009; Kageyama et al. 2014; Matsuo et al. 2001). In addition, greater peak extension velocity during the stride foot contact phase has been associated with greater ball speed in baseball pitching (Dowling ym. 2022; Kageyama et al. 2014).

6 METHODS

This chapter includes a description of the study setting as well as information about the participants. The measurements were conducted in the sports laboratory of the University of Jyväskylä during February and March 2025.

6.1 Participants

The participants were recruited using an open invitation on a local social media platform. The criteria for participation were to be an under 40-year-old adult male with an ability to throw the disc consistently over 110 meters with the right hand. The throwing distance was self-reported, and it was not confirmed before attending the study.

Ten participants (age 23,2 ± 2,5, height 183,7 ± 3,6 cm, body mass 86,2 ± 9,5 kg) who all fulfilled the criteria were included in the study. The competition experience of the participants ranged from hobby competitions to international elite level competitions. The participants were classified as professional or amateur based on their PDGA (Professional Disc Golf Association) status. Two participants had no PDGA status.

The participants were instructed to avoid heavy exercise 24 hours prior to the measurements and to only attend if they were in good health and injury-free. Before signing an informed consent document, the participants were given information about the study setting and the potential risks caused by participation. A description of how their personal data was processed was also given.

6.2 Study setting

The participants performed a 10–20 minute warm up of their own preference. The participants were allowed to throw submaximal throws during this time. One unnatural aspect of the study setting was a small step 1,5 meters before the force plate. Therefore, the beginning of the run-up area was approximately 5 centimeters above the ending of the run-up area and the force plate. Each participant was instructed to practice the run-up so that the step would not affect

their throwing. As a wired goniometer (details presented later) was used, the participants were allowed to perform more warm up throws after it was attached to get used to it and prepare for maximal performance.

During the actual measurements, each participant performed at least five successful throws. If the launch speed was greatest on the fifth trial, the participants were asked to continue throwing until the launch speed would no longer increase or until a limit of ten throws was achieved. The rest interval between each trial was at least one minute, however, the participants were otherwise allowed to determine when they were ready to throw. To be considered a successful throw, the lead foot had to be fully on the force plate with the rear foot not touching any part of the plate prior to disc release. If these conditions were not fulfilled, the participants were granted an extra trial. Other reasons for extra trials were the potential malfunctions of the measuring devices or if the participants felt that the trial had been negatively affected by the wired goniometer or the step. One throw for participants seven and ten had to be disqualified afterwards due to an incorrect foot position on the force plate. Their second-best trials were used for analysis.

6.3 Ground reaction forces

The ground reaction forces were measured using a 60 x 125 cm force plate (University of Jyväskylä, Finland) with a sampling rate of 1000 Hz. The data was processed on Signal software (version 4.10, Cambridge Electronic Design Ltd, Cambridge, UK), where the force plate data was transferred via an amplifier. The force plate was mounted to the ground and covered with a two-centimeter-thick tartan surface. The force plate measured forces in three directions that were vertical (Fz), anteroposterior (Fy) and mediolateral (Fx). The main directions of the reaction forces were upward, posterior (referred to as braking) and medial. The posterior direction was opposite to the movement direction, while the medial direction was normal to this, going from left to right. The posterior direction was negative, while the two other directions were positive. The medial direction may appear conflicting, as after the lead foot is rotated toward the target, the medial direction is pointing to the lateral side of the foot. However, during the action phase, the lead foot points away from the target, and therefore, it is more natural to refer to the left-to-right direction as medial. This is better illustrated in appendix 1, together with the orientations of the other two reaction force components.

Of the force plate data, the absolute peak vertical, braking and medial forces were analyzed. The relative forces were calculated by adjusting the peak values to body weight. The two horizontal components were used to calculate the resultant horizontal force (Frh). The function of this component was to cancel the inter-individual differences in stance deviation (Leppilahti 2019), and run-up orientation relative to the target. This is illustrated in appendix 1. Additionally, the total resultant force (Frt) was calculated using all three components.

The baseline signal of each force component was determined by calculating the mean value of the force plate data over a 0,5 second period prior to the foot contact. The force data was adjusted correctly based on these calculations. The body weight of the participants was measured on the force plate. The participants were asked to stand still on the force plate for ten seconds. The body weight was determined from the vertical force data, as an average of two one-second time intervals to minimize error. The baseline of the vertical force was adjusted as explained above. The body weight was rounded to the nearest integer. In the analysis, foot contact was determined based on the vertical force, as the point where 10 N was exceeded (McNally et al. 2015). The standard deviation of the vertical force baseline signal was less than 1,5 N, except for participant 6, which was approximately 7 N. To make sure that the foot contact had in fact occurred when the vertical force exceeded 10 N, the mediolateral force was also observed.

6.4 Knee kinematics

The knee flexion angle was measured with a goniometer (University of Jyväskylä, Finland) that was also connected to Signal software via an amplifier. The method used for calculating the knee flexion angle is illustrated in figure 7. As the goniometer had a four-meter-long wire, a research assistant had to keep it away from the participants' feet during the trials. Due to the orientation of the goniometer, knee flexion angle data was negative. The results are also expressed as negative values, zero representing no flexion and -50, for example, representing 50 degrees of flexion. Furthermore, positive knee extension velocity means that the knee is extending, while negative knee extension velocity indicates knee flexion.

The goniometer was attached to the lateral side of the lead leg, which was the right leg in all participants (figure 7). The pivot of the goniometer was set in between the lateral tibial condyle

and the lateral femoral condyle. The upper half of the goniometer was directed toward the trochanter major and the lower half to the lateral malleolus of the ankle. The upper and lower halves were attached to the skin using tape and two bands. The correct positioning and functioning were confirmed by measuring the angle when the knee was fully extended and in a squat position. This data was compared to measurements done with a manual goniometer simultaneously.

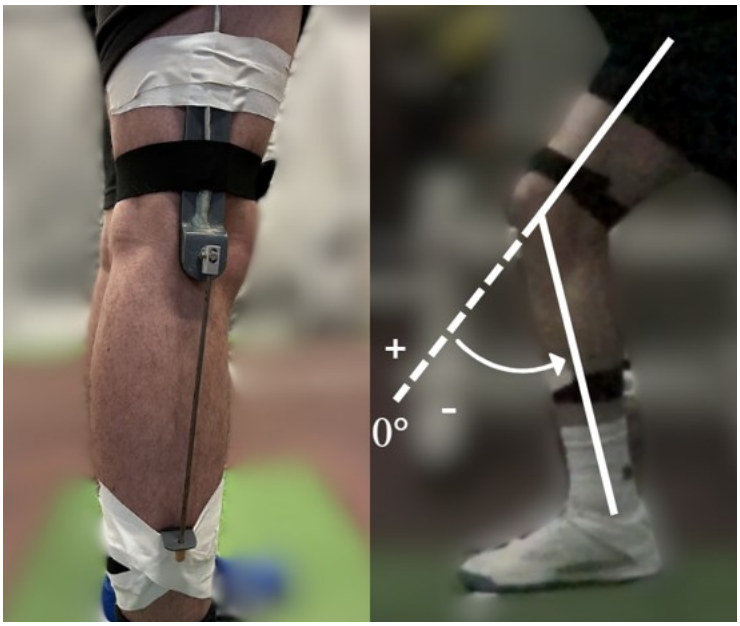

FIGURE 7. The position of the goniometer (left) and illustration of the knee flexion angle (right).

The knee extension velocity at each data point was calculated from the goniometer data by first dividing the change of angle by the change of time, which was 0,001 seconds. A moving average was calculated from this data. The average was calculated over a five-millisecond period consisting of values two milliseconds before and after the given time point. The peak extension velocity and the extension velocity at release were determined from this moving average data. The peak extension velocity was analyzed in two ways. In the first method, the analysis was limited to the disc release and the analysis began from the point where the extension velocity remained positive until the release. In the second method, the analysis included the follow-through phase until the extension velocity was no longer positive.

The timing of peak extension velocity was analyzed as percentage of time, foot contact representing 0 % and disc release representing 100 %. The same method was used for the timing

of peak forces. Net extension was calculated by subtracting the knee flexion angle at release from the flexion angle at foot contact. The knee movement patterns were categorized based on the extension velocity curves and further analyzed by visual inspection of the knee angle curves.

6.5 Launch speed and the timing of disc release

The launch speed was measured using a TechDisc (TechDisc inc., Overland Park, Kansas, USA), which is a disc with an attached accelerometer. The model used in the study was a Discraft Undertaker, which is a nine-speed driver disc, weighing 175 grams.

The trials were videotaped using a Sony RX10 II camera (Sony Group Corporation, Tokyo, Japan) with a frame rate of 250. The camera was set perpendicular to the direction of the movement and aligned with the center of the force plate. The videos were used to determine the moment when the disc was released. To synchronize the video and the force plate data, a light signal was set to go off when the vertical force exceeded 50 N on the force plate.

The disc was considered released when the thumb was no longer touching the top of the disc. Since the frame rate of the videos was 250, one frame covered a time of four milliseconds. The disc release was determined to have happened between the frame where the disc was visibly released and the preceding frame, minimizing potential error to two milliseconds. The timing of disc release was determined by calculating the time from the first frame where the light signal was visible to the instant of disc release. The light was on for 102 milliseconds. In nine videos, the light signal was visible in 26 frames (104 milliseconds) and in one video it was only visible in 25 frames (100 milliseconds), which was an additional error caused by the frame rate. This difference was considered and fixed, minimizing the potential error to one millisecond. In total, the potential error in the timing of disc release was ± 1–3 milliseconds.

The videos were also used to ensure that the launch speed given by the TechDisc was reliable. The best throw of each participant was analyzed on Kinovea (version 2023.1.2, Free Software Foundation Inc., Boston, Massachusetts, USA). A 111,5 x 155,5 cm framework was used to calibrate the videos. The two-dimensional motion analysis resulted in a maximum difference of 5 % for each launch speed compared to the TechDisc. Therefore, the values measured with the TechDisc were acceptable to be used in the analysis.

6.6 Statistical analysis

The statistical analysis was performed on SPSS (version 28.0.0.1, International Business Machines Corporation, Armonk, New York, USA) and Microsoft Excel 2025 (version 2502, Microsoft Corporation, Redmond, Washington, USA). The normality of distribution for each variable was assessed using a Shapiro-Wilk test. Absolute peak medial force and time to peak vertical force were not normally distributed. Therefore, their relationship with launch speed was tested using Spearman's rank correlation. Pearson’s product correlation test was used for the rest of the variables. The two-tailed significance level was set at $p < 0,05$.

For each force and knee kinematic variable, the mean and standard deviation, as well as the minimum and maximum values are presented. Additionally, the launch speed and length of the action phase are presented individually for each participant. The participants were put in order based on launch speed. P1 refers to the participant with the highest launch speed and P10 refers to the participant with the lowest launch speed.

7 RESULTS

The average launch speed was 30,7 ± 2,7 m/s and the average length of the action phase was 0,199 ± 0,028 seconds. The highest launch speed was 35,8 m/s, while the lowest was 27,1 m/s. Individual results are presented in table 1.

TABLE 1. Launch speed and length of the action phase.

Participant	Launch speed (m/s)	Action phase length (s) [a]	Classification [b]
P1	35,8	0,149	Professional
P2	33,3	0,186	Professional
P3	32,9	0,167	Amateur
P4	31,2	0,203	- [c]
P5	30,9	0,239	Amateur
P6	30,2	0,189	Amateur
P7	29,0	0,219	Amateur
P8	28,9	0,193	Professional
P9	27,4	0,221	- [c]
P10	27,1	0,227	Amateur

[a] Time from foot contact to disc release, [b] Based on PDGA status, [c] No PDGA status.

7.1 Force parameters

The absolute and relative peak forces are presented in table 2. A correlation with launch speed was observed in five peak force parameters that were the absolute peak braking force (r = -0,748, p = 0,013), the relative peak braking force (r = -0,735, p = 0,015), the absolute peak horizontal resultant force (r = 0,732, p = 0,016), the relative peak horizontal resultant force (r = 0,734, p = 0,016) and the absolute total resultant force (r = 0,669, p = 0,034). The statistically significant correlations are illustrated in figures 8, 9 and 10.

The temporal force parameters are presented in table 3. None of the parameters had a statistically significant correlation with launch speed. On average, the peak braking and peak medial forces were achieved quicker than the peak vertical force.

TABLE 2. The absolute and relative peak forces and their correlation with launch speed.

Parameter	Mean	SD [b]	Min	Max
Peak vertical force (N)	1584	423	1078	2494
Peak braking force (N)	-925	395	-1783	-516
Peak medial force (N)	621	196	434	1106
Peak horizontal resultant force (N)	1048	411	597	2002
Peak total resultant force (N)	1873	571	1227	3168
Peak vertical force (BW) [a]	1,89	0,51	1,26	2,67
Peak braking force (BW)	-1,10	0,45	-1,91	-0,60
Peak medial force (BW)	0,74	0,22	0,51	1,18
Peak horizontal resultant force (BW)	1,25	0,46	0,70	2,15
Peak total resultant force (BW)	2,24	0,67	1,43	3,40

[a] BW, force relative to body weight, [b] Standard deviation.

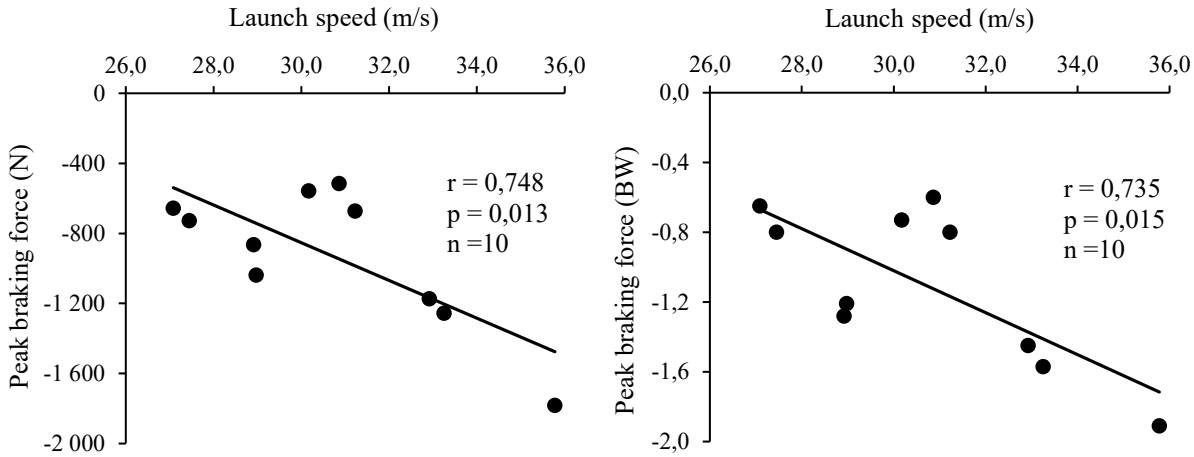

FIGURE 8. The correlation between launch speed and the absolute and relative (BW) peak braking forces.

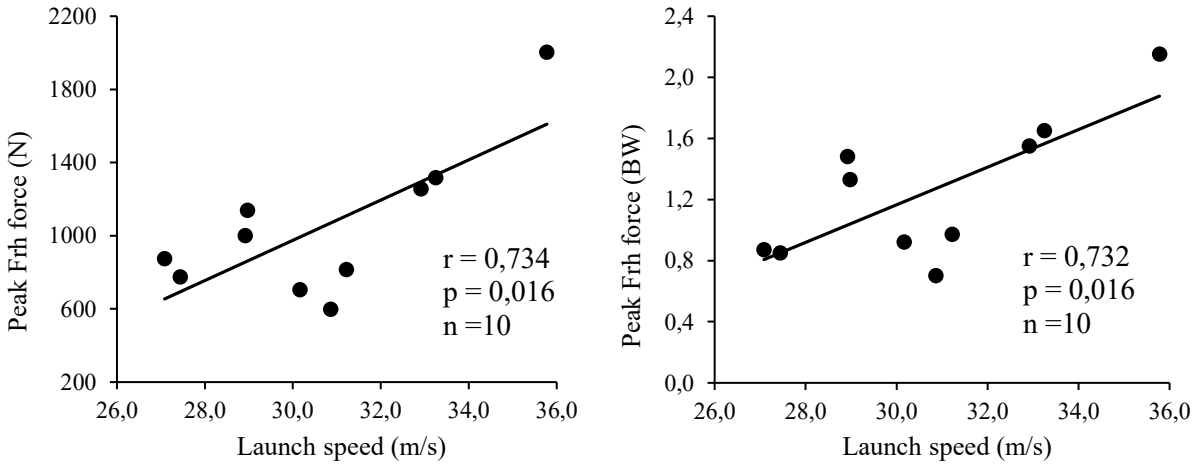

FIGURE 9. The correlation between launch speed and the absolute and relative (BW) peak horizontal resultant forces (Frh).

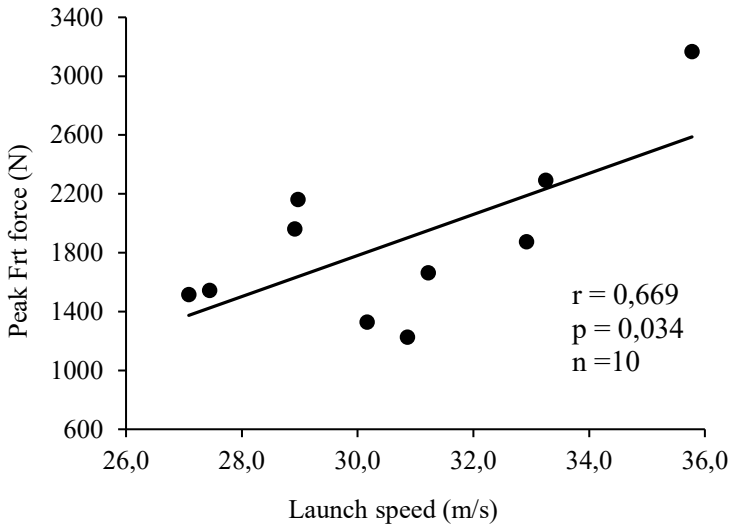

FIGURE 10. The correlation between launch speed and the absolute peak total resultant force (Frt).

TABLE 3. The temporal force parameters.

Parameter	Mean	SD [a]	Min	Max
Time to peak vertical force (s)	0,11	0,04	0,07	0,20
Time to peak braking force (s)	0,09	0,04	0,05	0,18
Time to peak medial force (s)	0,09	0,04	0,01	0,15
Time to peak horizontal resultant force (s)	0,09	0,03	0,04	0,14
Time to peak total resultant force (s)	0,09	0,03	0,04	0,14
Timing of peak vertical force (%)	54,5	17,6	29,9	89,9
Timing of peak braking force (%)	46,6	14,7	28,1	80,2
Timing of peak medial force (%)	44,9	16,5	7,2	64,0
Timing of peak horizontal resultant force (%)	46,4	9,6	34,1	65,2
Timing of peak total resultant force (%)	45,8	12,1	26,4	61,4

[a] Standard deviation.

Examples of the vertical (Fz), anteroposterior (Fy) and mediolateral (Fx) force curves for P1 are presented in figure 11. A similar graph for each participant is presented in appendix 2. The mediolateral force had the most visible impact peak of the three components. Its magnitude decreased after the impact peak and increased again rapidly. Its magnitude decreased toward the end of the action phase. The vertical and anteroposterior forces reached their peak values halfway through the action phase. The vertical force decreased to the level of BW at release.

The magnitude of the braking force also decreased linearly during the latter half of the action phase. No impact peak was observed in the anteroposterior force, as it remained near zero briefly after foot contact.

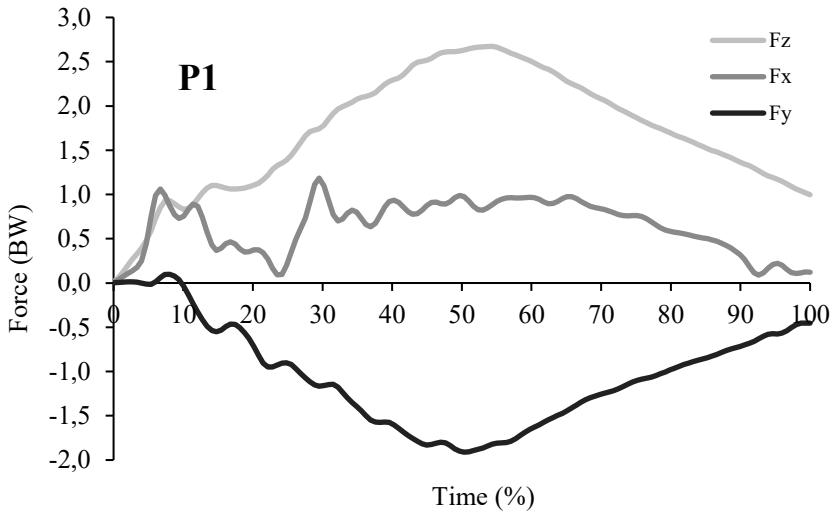

FIGURE 11. Vertical (Fz), mediolateral (Fx) and anteroposterior (Fy) force curves of P1 relative to body weight (BW). The X-axis represents the time from foot contact to disc release.

The force curves varied among the participants. The vertical force had a tendency to include at least two peaks. The first peak was the greatest in magnitude except for P10, who reached the peak force just prior to disc release. The anteroposterior force had a similar trend to the vertical force as it could be described as a reflection of the vertical force, to some extent. For most participants, the anteroposterior force remained near zero briefly after foot contact, which was followed by an increase toward the peak force, similar to P1. P5 and P10 differed from the rest as the magnitude of the braking force did not start to increase until after the first third of the action phase. The mediolateral force curves were generally similar to P1, while the magnitude of the first impact peak varied. Its direction changed from medial to lateral for three participants close to disc release. The total resultant force curves together with knee angle curves are presented in appendix 3. An impact peak is visible for participants who were extending their knee at foot contact, but not for the participants who approached foot contact with a flexing knee.

7.2 Knee kinematics

The kinematic variables of the lead knee are presented in table 4. Peak extension velocity was correlated with launch speed (r = 0,781, p = 0,008) when the follow-through phase was included in the analysis, but not when the analysis was limited to disc release. None of the other parameters were correlated with launch speed. The statistically significant correlation is illustrated in figure 12.

TABLE 4. Knee kinematic parameters and their correlation with launch speed.

Parameter	Mean	SD [c]	Min	Max
Peak extension velocity (°/s) [a]	557,9	198,0	323,5	848,4
Peak extension velocity (°/s) [b]	495,9	188,4	262,5	848,4
Extension velocity at release (°/s)	376,0	137,8	225,5	603,6
Timing of peak extension velocity (%) [a]	96,9	21,1	64,4	130,7
Timing of peak extension velocity (%) [b]	87,0	12,4	64,4	100,0
Net extension (°)	8,2	15,6	-22,4	27,0

[a] Analysis includes the follow-through phase, [b] Analysis ends at disc release, [c] Standard deviation.

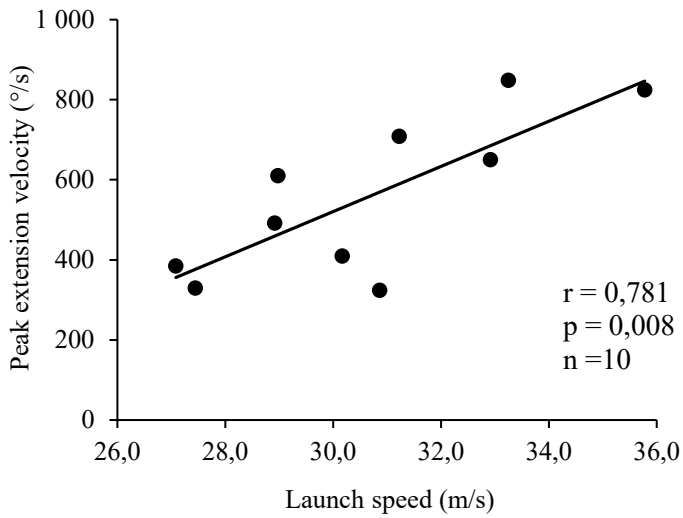

FIGURE 12. The correlation between launch speed and peak knee extension velocity when the follow-through phase was included in the analysis.

The knee movement patterns were classified into categories 1, 2 and 3. Based on the extension velocity curves (appendix 4) and knee angle data, pattern 1 (figure 13) begins with extension. The extension is followed by two flexion-extension patterns and therefore the knee is extending at release. This pattern was observed in P2, P7 and P9. Pattern 2 (figure 14) can be defined as a subgroup of pattern 1. This pattern was identified in P1, P3, P6 and P10. The pattern is similar to pattern 1, but the second flexion is minimal, as it was limited to 1,5 degrees of angle change in all participants. P10 differed from the rest in pattern 2, as his net knee extension was negative.

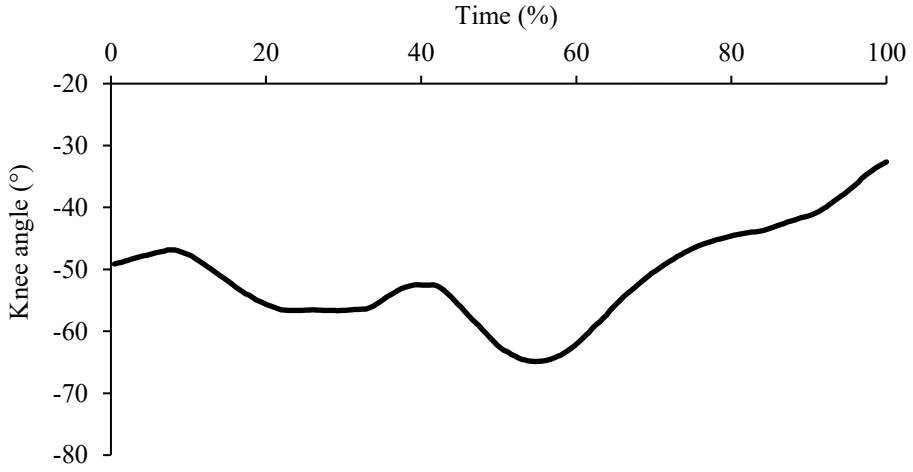

FIGURE 13. Knee movement pattern 1 during the action phase. The X-axis represents the time from foot contact to disc release.

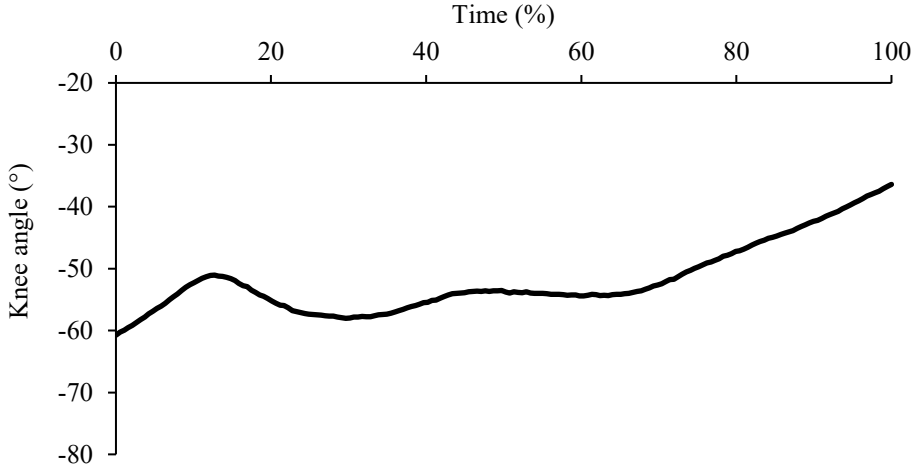

FIGURE 14. Knee movement pattern 2 during the action phase. The X-axis represents the time from foot contact to disc release.

Movement pattern 3 is presented in figure 15. This pattern begins with flexion or with minimal change in knee angle. The first half of the action phase consists of flexion, which is followed by continuous extension until release. This was observed in P4 and P5. P8 could not be classified with certainty into any category, but the pattern was closest to the definition of pattern 3 based on the extension velocity curve.

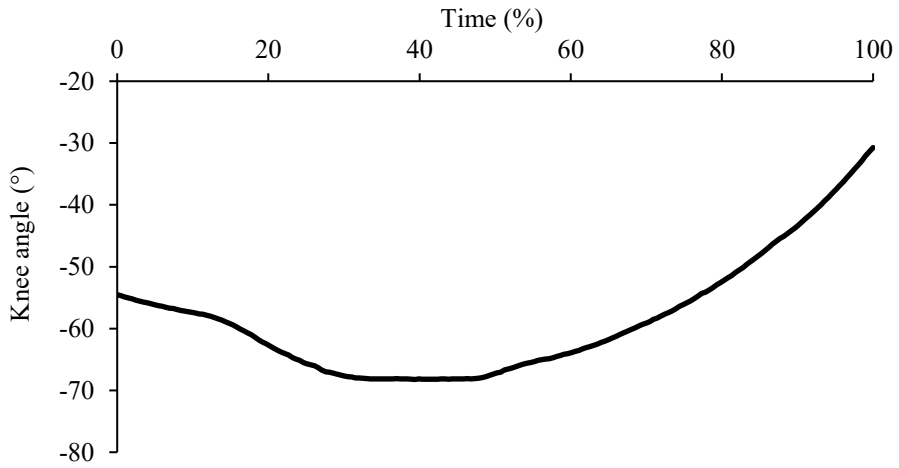

FIGURE 15. Knee movement pattern 3 during the action phase. The X-axis represents the time from foot contact to disc release.

8 DISCUSSION

A correlation with launch speed was observed in the absolute and relative peak braking and horizontal resultant forces, as well as in the absolute total resultant force. Of the knee kinematic parameters, peak extension velocity had a correlation with launch speed when the follow-through phase was included. Additionally, three knee movement pattern categories were identified and described.

8.1 Launch speed

The highest launch speed (35,8 m/s) and the group average (30,7 ± 2,7 m/s) were significantly lower than in the study by Leppilahti (2019) (41,2 m/s and 36,4 m/s, respectively). The lower average in the current study can be explained by the lower participation criterion considering the self-reported throwing distance. However, neither study confirmed the throwing distance prior to attending, so precise comparison cannot be made. The difference in the highest launch speed is surprising, as the throwing performance of P1 in the current study can be considered elite level. The best performing participant in Leppilahti's (2019) study was described similarly. A few methodological differences could partly explain this difference. The disc used in the current study could be considered a fairway driver or a slow-speed distance driver, whereas Leppilahti (2019) used a high-speed distance driver. Additionally, the launch speed was measured with a TechDisc in the current study, whereas Leppilahti (2019) used a three-dimensional motion analysis. The TechDisc is yet to be validated in research use, although it has been used in the study by Koch et al. (2024). The launch speed values measured with the TechDisc were confirmed with a two-dimensional motion analysis, although this method is not fully reliable either, as the disc moves three-dimensionally during the action phase.

No statistical testing was performed considering launch speed and the classification of the participants. The participants were classified based on their possible PDGA status. This classification alone does not determine the skill level of the participants, although it works as an indicator. The two highest launch speeds were produced by professional level participants. However, the third highest launch speed was produced by a participant with very limited competition experience, while the third professional participant had one of the lowest launch

speeds of the participants. These results highlight the complexity of disc golf, as launch speed alone does not determine the overall performance level.

8.2 Peak ground reaction forces

Greater braking forces were associated with higher launch speed. This finding is in line with study results from baseball pitching (Kageyama et al. 2014; McNally et al. 2015; Wasserberg & Giordano 2023), javelin throwing (Reinholm 2024) and cricket bowling (Portus et al. 2004). In addition, peak horizontal resultant force was correlated with launch speed. The horizontal resultant force has a similar function to the braking force, but it includes the mediolateral force component. The braking forces have likely contributed more to the peak horizontal resultant force, as they were greater than the medial forces in general. Finally, greater relative total resultant force has been associated with better pitching performance in baseball (Kageyama et al. 2014; McNally et al. 2015), but in the current study, only the absolute peak force had a statistically significant correlation with launch speed. This would suggest that greater ground reaction forces would be beneficial for throwing performance regardless of body weight. However, conclusions based on this result should be drawn with caution, as the correlation of the relative peak force ($r = 0{,}627$, $p = 0{,}053$) was close to statistical significance.

Vertical and medial peak forces were not associated with launch speed. Greater peak vertical force has been associated with better throwing performance in baseball pitching (Kageyama et al. 2014; McNally et al. 2015), but its role seems to be less significant in the backhand throw based on the current results. However, the p-value of the correlation between absolute peak vertical force and launch speed was 0,055, which is again close to statistical significance. Therefore, its potential contribution to better performance should not be excluded. Interestingly however, the relative peak vertical force was further away from statistical significance. Although the peak medial force did not correlate with launch speed, its significantly greater magnitude compared to baseball pitching (Kageyama et al. 2014; McNally et al. 2015) suggests that it plays a more significant role in the backhand throw. This is likely due to the characteristics of the lead foot orientation and actions following foot contact. On the other hand, the magnitudes of the other peak forces were close to baseball pitching. Generally, the peak braking force was slightly greater, whereas the peak vertical and total resultant forces were slightly lower in the current study when compared to the results of Kageyama et al. (2014),

McNally et al. (2015), and Wasserberg and Giordano (2023). However, there was plenty of variation between the participants in the current study.

Wasserberg and Giordano (2023) have suggested that an increased braking force and the ability to withstand it could contribute to greater pelvis rotational velocity and, overall, allow better energy transfer through the kinetic chain. This could also apply to the backhand throw, although the angular velocity of pelvis rotation itself has not been associated with higher launch speed (Leppilahti 2019). According to Worthington et al. (2013b), greater braking forces can be partly explained by a quicker run-up speed. Furthermore, a quicker run-up speed has been associated with higher ball speed in cricket bowling (Worthington et al. 2013a). The run-up speed was not analyzed in the current study, so its significance remains a question. However, Worthington et al. (2013a) have suggested that there is likely an optimal run-up speed, as too quick of a run-up could cause an inability to maintain a sufficient technique. For example, Worthington et al. (2013b) have suggested that high ground reaction forces could cause increased knee flexion after foot contact if the forces cannot be withstood. Meanwhile, maintaining a straighter knee throughout the foot contact phase allows the linear momentum to be transferred into an efficient trunk rotation (Worthington et al. 2013a).

8.3 Temporal force parameters

No correlation between launch speed and the temporal force parameters was observed. Time to peak force had plenty of variation among the participants. In general, the time to peak force was greater than in cricket bowling (Portus et al. 2004; Worthington et al. 2013b) and in javelin throwing (Reinholm 2024). The peak forces observed in these studies were achieved quickly after the foot contact, and therefore, they can be classified as impact forces (Nigg 2007, 329). The impact forces in the current study were relatively small, which could be partly explained by the nature of the foot contact. The foot contact occurred toes first, whereas in cricket bowling and javelin throwing the foot contact occurs heel first (Reinholm 2024; Worthington et al. 2013b).

The relative timing of peak forces was generally earlier than in baseball pitching. On average, the peak vertical, braking and resultant forces were achieved at 45–55 % of the action phase, whereas Kageyama et al. (2014) reported values of approximately 70–88 % for the same

variables. These results are comparable considering the time from foot contact to release, which was almost identical on average. However, in the current study, particularly the timing of peak vertical and braking forces had plenty of variation, ranging approximately from 28 to 90 %. Therefore. making the comparison based on the group average is questionable. A later occurrence of peak vertical and resultant forces characterized the high-velocity pitchers in the study by Kageyama et al. (2014), but no such association was observed in the current study.

P10 had the longest time to peak vertical and peak braking forces. In fact, the time was almost twice the group average for both components. The peak magnitudes were among the lowest values observed, however, in line with the results of the other participants. The long time to peak forces would suggest inability to properly stop the linear momentum after foot contact. One explanation for this could be excessive knee flexion after foot contact, which P10 had the second highest amount among the participants. Of the participants, P5 flexed the knee the most after foot contact and, in fact, he had the second longest time to peak braking force. However, as these results are based on only one trial and no statistical analysis was performed, conclusions should be drawn with caution.

8.4 Characteristics of the forces

The vertical and anteroposterior forces could be defined as reflections of each other. Most participants had two visible peaks, especially in vertical force. On the other hand, the vertical and anteroposterior force curves of P1 could be described as a linear increase toward peak force, followed by a linear decrease. The vertical movement of the center of mass has not been studied in disc golf, so it is difficult to determine whether a linear acceleration is better than a more varying acceleration. The mediolateral force curves were different compared to the vertical and anteroposterior force curves. Common for all participants was the decreasing magnitude of the medial force toward disc release. This decrease is most likely due to the weight shifting on the lead foot from the toes to the heel in preparation of the leg rotation. Near disc release, three participants had a slightly negative mediolateral force. Leppilahti (2019) mentioned that one participant in his study did not rotate the lead leg after the disc release and in contrast to this, in the current study, P5 had fully rotated the leg prior to disc release. This early rotation could explain why the mediolateral force went negative, indicating a shift to laterally directed force. Leppilahti (2019) suggested that rotating the leg allows proper body rotation after the release

without causing excess stress to the ankle and the knee. However, the optimal timing of the leg rotation remains a question.

The mediolateral force tended to show a significant impact peak, while the anteroposterior force remained near zero briefly after foot contact. This would actually suggest, that at foot contact, force is mostly produced in the side-to-side direction. Furthermore, the linear momentum would be first slowed down in the medial direction as opposed to the posterior direction. This is likely a consequence of actions that also determine the amount of stance deviation. As the foot contact is approached, the lead knee experiences a slight extension, which moves the foot laterally in relation to the movement direction. In appendix 1, these actions appear as the lead foot being positioned more to the left than the rear foot. As the foot contact occurs, the knee extension causes a medially directed reaction force. The posteriorly directed reaction force starts to increase 15 to 20 milliseconds after the foot contact, likely as a consequence of a larger part of the foot contacting the ground. However, these descriptions should be viewed critically.

The significance and characteristics of the mediolateral and braking forces should be further investigated by controlling the effect of run-up orientation and stance deviation. Especially the mediolateral force could be an important factor for better throwing performance, although such result was not statistically confirmed in the current study. For example, some participants experienced a rapid increase in medial force as the knee was extended after the first flexion movement following foot contact. In conclusion, the mediolateral force could contribute significantly to knee extension early in the action phase.

8.5 Knee extension velocity

The peak extension velocity was analyzed using two methods. No correlation with launch speed was observed when the analysis was limited to disc release. However, when the follow-through phase was included, the correlation was statistically significant ($r = 0{,}781$, $p = 0{,}008$). Matsuo et al. (2001) have suggested that higher extension velocity during the pitch stabilizes the lead knee better which could allow a more efficient trunk rotation in baseball pitching. A similar effect is possible in the disc golf backhand throw. Four participants achieved the peak extension velocity during the follow-through phase. P1 and P2 had an increasing extension velocity at release and the peak followed soon after. For P6 and P10, the extension velocity was decreasing,

although positive, at release but another peak followed. Overall, there was a tendency for multiple peaks in extension velocity throughout the action phase and the follow-through phase.

The peak extension velocities were generally greater than in baseball pitching (Dowling et al. 2022; Kageyama et al. 2014; Matsuo et al. 2001). The nature of the backhand throw may allow greater knee extension velocity to be produced compared to baseball pitching. However, reliable comparison cannot be made, as the measuring and analysis techniques were different. In the current study, the peak extension velocity occurred at 87,0 % or at 96,9 % of the action phase on average, depending on the analysis method. The results of Chu et al. (2009) and Kageyama et al. (2014) differed significantly from each other. Comparison is even more difficult due to the use of relative timing, as the foot contact phases varied in absolute length. The peak extension velocity occurred significantly later in the current study compared to Kageyama et al. (2014) but earlier compared to Chu et al. (2009). No association between launch speed and the timing of peak extension velocity was observed, contradictory to Kageyama et al. (2014).

Knee extension velocity at release did not correlate with launch speed. Its significance in the backhand throw remains unclear even though a higher extension velocity at release has been associated with better performing baseball pitchers (Chu et al. 2009; Kageyama et al. 2014; Matsuo et al. 2001). The extension velocity at release is affected by the timing of peak extension velocity, which had plenty of variation among the participants. Some participants had already reached the peak velocity prior to the release and the extension velocity had therefore begun to decrease. For example, of the participants, P2 achieved the highest extension velocity, which occurred at 84 % of the action phase. However, the extension velocity at release was the fourth lowest. On the other hand, as some participants reached the peak velocity soon after release, the extension velocity was increasing at release. As the extension velocity at release varied among the participants due to the reasons presented above, it is difficult to analyze its contribution to throwing performance. It should also be noted that, as the determination of the disc release had a potential error of a few milliseconds and the knee extension velocity changed significantly between each data point, the use of statistical testing is questionable.

Most participants had a positive knee extension velocity from 60–65 % of the action phase through release (appendix 4). Interestingly, P1 and P2 were exceptions as the extension velocity of P2 turned positive much later, at 74 %. P1 on the other hand, had a positive extension velocity

starting from 27 %, which was significantly earlier than the rest. However, the extension velocity remained near zero from 72 to 84 %, while being negative occasionally. However, these single negative extension velocities were caused by flexion that was almost non-existent and could be explained by deviation in the signal. None of the participants had a negative extension velocity at release, meaning that everyone was extending their knee through release. As the number of participants was small, it cannot be determined whether this is natural in the backhand throw or if the participants represented a good technique. In contrast, a flexing knee through release has been classified as a factor contributing negatively to pitching performance in baseball (Chu et al. 2009)

Knee kinematics received little attention in the kinematic analysis of the backhand throw by Leppilahti (2019). As a more stable knee and greater knee extension velocity have been suggested to contribute to better trunk rotation (Matsuo et al. 2001; Worthington et al. 2013a), these relationships should be investigated in disc golf. Especially the contribution of peak knee extension velocity to launch speed remains a question, as some players reached it after disc release, which in baseball pitching has been considered as an insufficient technique (Chu et al. 2009). As the peak angular velocities of pelvis and trunk rotation should occur prior to disc release (Leppilahti 2019), the knee extension velocity after disc release likely does not contribute to these variables. However, the peak extension velocity, regardless of it possibly occurring after disc release, seems to play a role in producing a greater launch speed based on the current results.

8.6 Knee movement patterns

The knee movement patterns were classified into three categories based on the extension velocity curves and further analyzed based on visual inspection of the knee angle curves. Movement patterns 1 and 2 were similar except for the second flexion seen in pattern 1. Instead of a visible flexion movement, the knee angle stayed almost unchanged for a short time period in pattern 2. The categories are not definite as the timing and amount of change in knee angle varied among the participants within the same category. Due to the low number of participants, statistical analysis considering the movement patterns was not possible. Interestingly, pattern 1 was observed in participants with the two lowest launch speeds and the participant with the second highest launch speed. Similarly, pattern 2 was recognized in participants with the

highest and lowest launch speed. Although, including P10 in pattern category 2 is questionable, as he had significantly more flexion after foot contact compared to the rest in the same category. He was also one of the two participants, who had more knee flexion at release than at foot contact. However, net extension did not correlate with launch speed, in contrast to the results of Dowling et al. (2022) in baseball pitching.

The first knee flexion movement following the foot contact is likely due to the impact forces, as knee flexion reduces their magnitude (Portus et al. 2004). The impact forces were relatively small in the current study, but knee flexion could still have been used to absorb them. The second flexion in pattern 1 could be due to rapid weight shifting toward the lead leg. On the other hand, participants in pattern 2 were possibly able to keep extending the knee regardless of the weight shift, thus allowing proper pelvis and trunk rotation (Worthington et al. 2013a), for example. Therefore, the knee movements should be investigated in relation to the rest of the kinetic chain.

Pattern 3 included constant flexion followed by constant extension. P4 and P5, whose lead knee was flexing at foot contact, had the smallest impact forces, but their launch speed was among the better half of the participants. In conclusion, it is difficult to determine whether one knee movement pattern is better. However, it seems that the lead knee tends to move in a pattern that includes alternating flexion and extension movements regardless of the throwing performance level. Future studies should specify the categorization of the patterns, as the use of extension velocity may result in unnecessarily broad criteria that do not distinguish the patterns well enough.

Similarities in knee movement patterns were observed when compared to javelin throwing (Whiting et al. 1991) and baseball pitching (Matsuo et al. 2001). Pattern 1 had a double flexion-extension pattern. However, it is not identical to the one observed in javelin and baseball as it had an additional extension phase at the beginning that was not seen in the studies by Whiting et al. (2001) and Matsuo et al. (2001). Pattern 2 itself is not comparable to the patterns presented by Matsuo et al. (2001), although as it was identified as a subgroup of pattern 1 in the current study, it can be defined as closest to the double flexion-extension pattern. Pattern 3 on the other hand, is similar to type B observed in baseball pitching (Matsuo et al. 2001) as it consisted of a continuous flexion phase followed by a continuous extension through release. As each participant in the current study was extending their knee through release, the pattern types

associated with weaker throwing performance in the study by Matsuo et al. (2001) were not observed.

If the classification was done based on the criteria of Portus et al. (2004), eight participants would have been flexor-extenders and two participants extenders. These two categories were generally associated with higher ball speed in the study by Portus et al. (2004). Similarly, only the patterns that Matsuo et al. (2001) associated with high-velocity pitchers could be somewhat observed in the current study. In simple terms, this would suggest that the participants in the current study represented a good technique. However, this conclusion should not be made. As the current study was the first one to investigate the movement patterns and the number of participants was low, it is likely that multiple different patterns exist. The mechanics seem to have similarities with baseball pitching and javelin throwing, but analysis of optimal technique based on the optimal technique in other overhead throwing motions may not be reasonable in a disc golf backhand.

8.7 Limitations

The current study has multiple limitations. The correlation test results were likely affected by the low number of participants. In particular, P1 stood out from the rest considering the peak force parameters and launch speed, which can be seen in the scatter plots in the results section. This may have caused distortion considering the strength of the correlations. Furthermore, the results were based on only one trial per participant. As the backhand throw is a complex movement, the trials of each participant have most likely not been identical. For example, analyzing the second-best trial instead of the best trial of each participant could have led to significantly different results. Additionally, some correlations were close to statistical significance, and their potential contribution to higher launch speed should not be excluded.

The measuring devices could have caused some error in the results. The goniometer was attached with bands and tape, but possible excess movement during the trials could have occurred. Its orientation could have had an error of multiple degrees as it was positioned based on palpation of the anatomical structures. Due to these reasons, knee angle at particular time points was not reasonable to analyze. Use of a three-dimensional motion analysis is suggested for future studies. The second limitation has to do with filtering. The force plate data was not

filtered. This could have influenced the true peak values. The data was smooth in general, as the standard deviation of the baseline signal was less than 1,5 N. However, the baseline signal of the vertical force data of P6 had a standard deviation of approximately 7 N. Therefore, the peak forces could have differed a few dozen newtons from the true values. Similarly, the goniometer data was not filtered. However, the extension velocity data was filtered with a five-millisecond moving average, which evened out the data and slightly reduced the magnitudes of the highest peaks.

The run-up orientation or greater stance deviation may have affected the magnitudes of the anteroposterior and mediolateral forces. For example, a diagonally oriented run-up causes a decrease in the braking force and an increase in the mediolateral force compared to a straight run-up. This is due to the functioning of the force plate. This effect could not be canceled in the current study. However, the horizontal resultant force was created to analyze the true magnitude of the force produced to stop the linear momentum, canceling the effect of run-up orientation. Therefore, the result considering the importance of horizontally produced force is relevant, while the true magnitude of each horizontal force component may not be exact.

The performance of the participants could have been negatively affected by the wire of the goniometer and the step prior to the force plate. However, none of the participants asked for an extra trial that would have been granted if these factors had disturbed the trial. The beginning of the run-up area was approximately five centimeters above the level of the force plate, which could have theoretically affected the run-up speed and therefore the forces. However, this effect is believed to be minimal.

The timing of disc release was calculated based on the videos. The error caused by the frame rate was minimized to the best extent. The disc was considered released when the thumb was no longer touching the disc. This moment could be determined clearly in most videos. In some videos, the thumb could be seen being on the limit of not touching the disc anymore, but as it could not be confirmed, the disc was not considered released until the following frame. This could have possibly caused more error in the timing of the disc release. However, each controversial case was solved using the same method and therefore the direction of the error was always the same.

The final major limitation is that the analysis was based on the launch speed. The use of launch speed included a presumption that higher speeds are an indication of a better technique. However, launch speed alone does not determine the final length of a throw. Furthermore, disc golf requires accuracy, which was not considered in the current study.

The participant group consisted of amateur and professional male disc golfers. The amateur participants can be considered intermediate to advanced level players, while two of the professional participants were elite level. Future studies should include participants with a lower throwing performance as well as female players, as these two groups are yet to receive attention in studies considering throwing kinetics and kinematics.

8.8 Conclusion

The hypotheses can be mostly confirmed. The results indicate that being able to create greater ground reaction forces, particularly in the posterior direction, may contribute to a higher launch speed. The results also suggest that greater lead knee extension velocity is associated with higher launch speed, although its contribution remains a question as the timing of peak values varied significantly. However, having a greater knee extension velocity at release does not appear to contribute to a higher launch speed. Finally, this study provided a description of different knee movement patterns observed during the throw, but no statistical analysis was performed.

The magnitudes of the ground reaction forces were close to baseball pitching except for the medial force, which was significantly greater in the backhand throw. The knee kinematics have some similarities across baseball pitching, javelin throwing and the disc golf backhand throw. However, the current results suggest that analysis of an optimal technique in disc golf based on the findings from javelin throwing and baseball pitching may not be reasonable.

Future studies should investigate the effect of run-up speed on ground reaction forces and launch speed. Knee kinematics should be further investigated together with the kinematics of the pelvis and the following segments of the kinetic chain. In particular, the effect of the timing of peak extension velocity on the timing of other segmental motions should be investigated.

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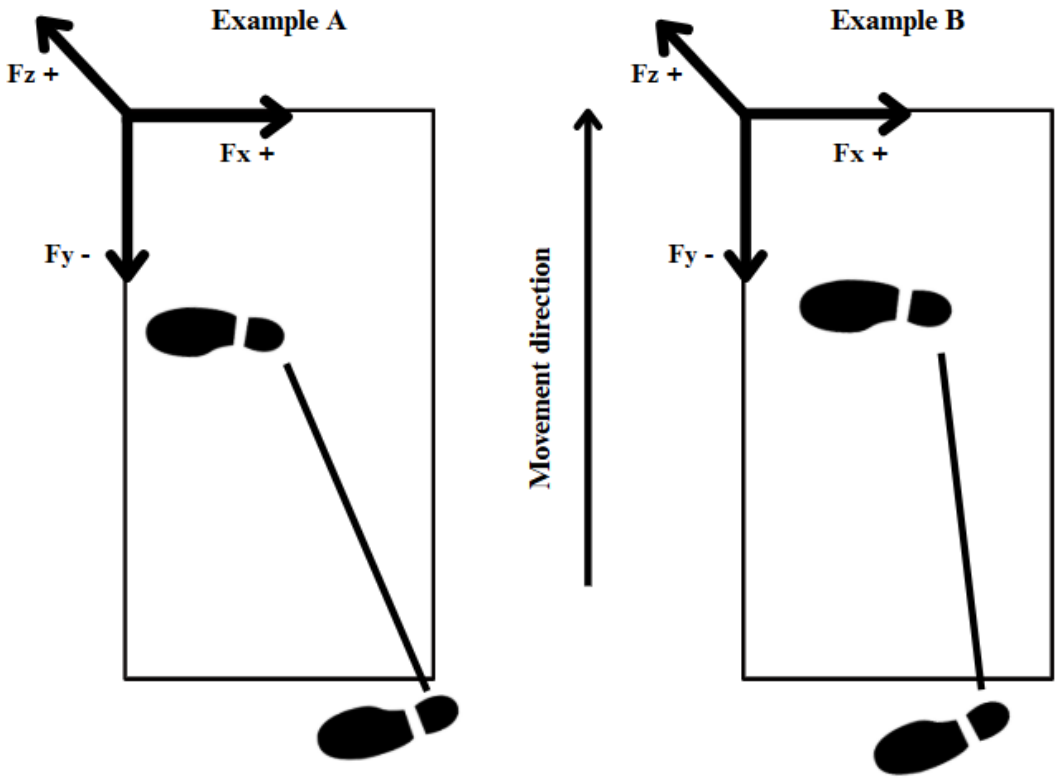
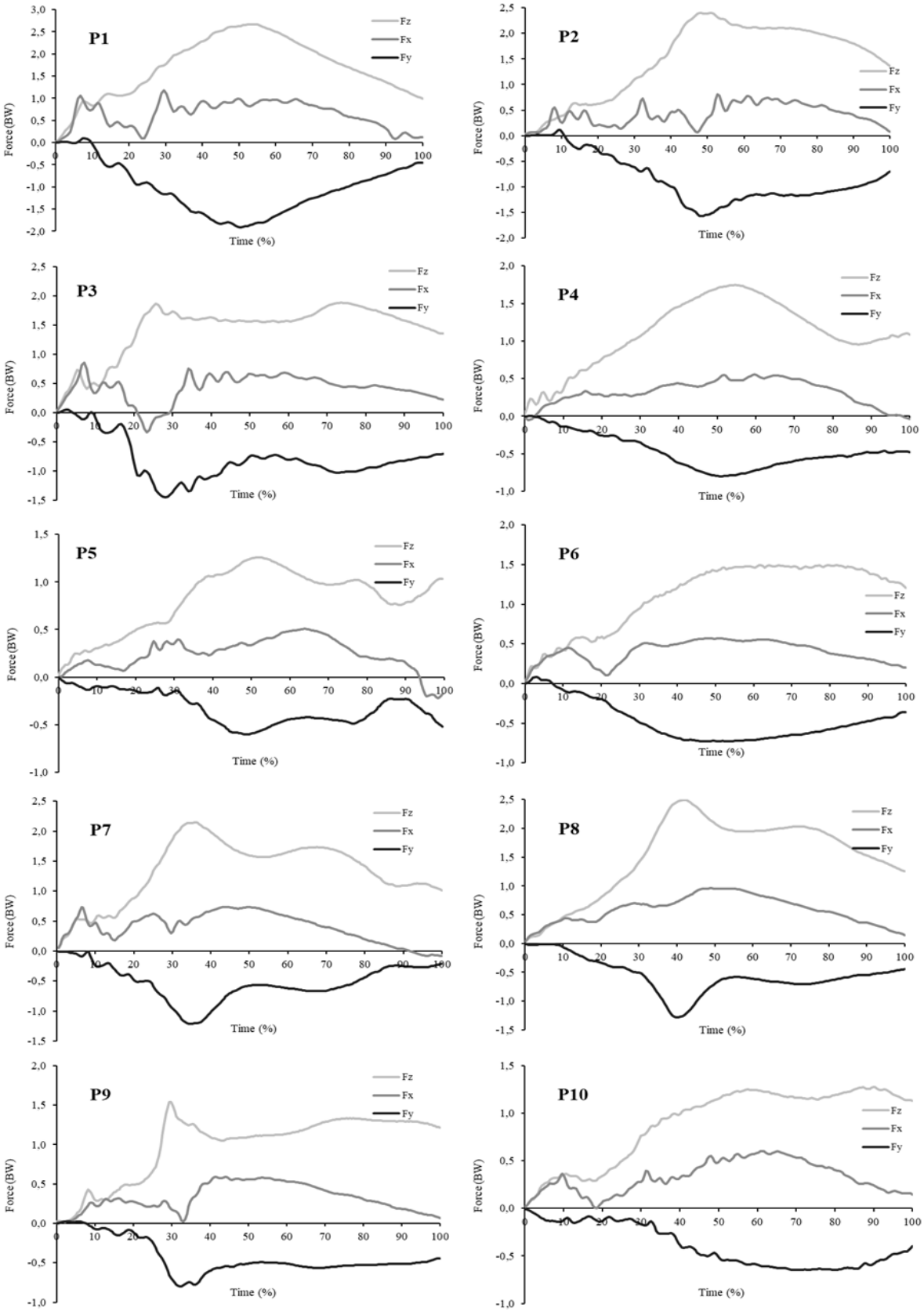
APPENDIX 1. Run-up orientation and stance deviation examples during the last step.

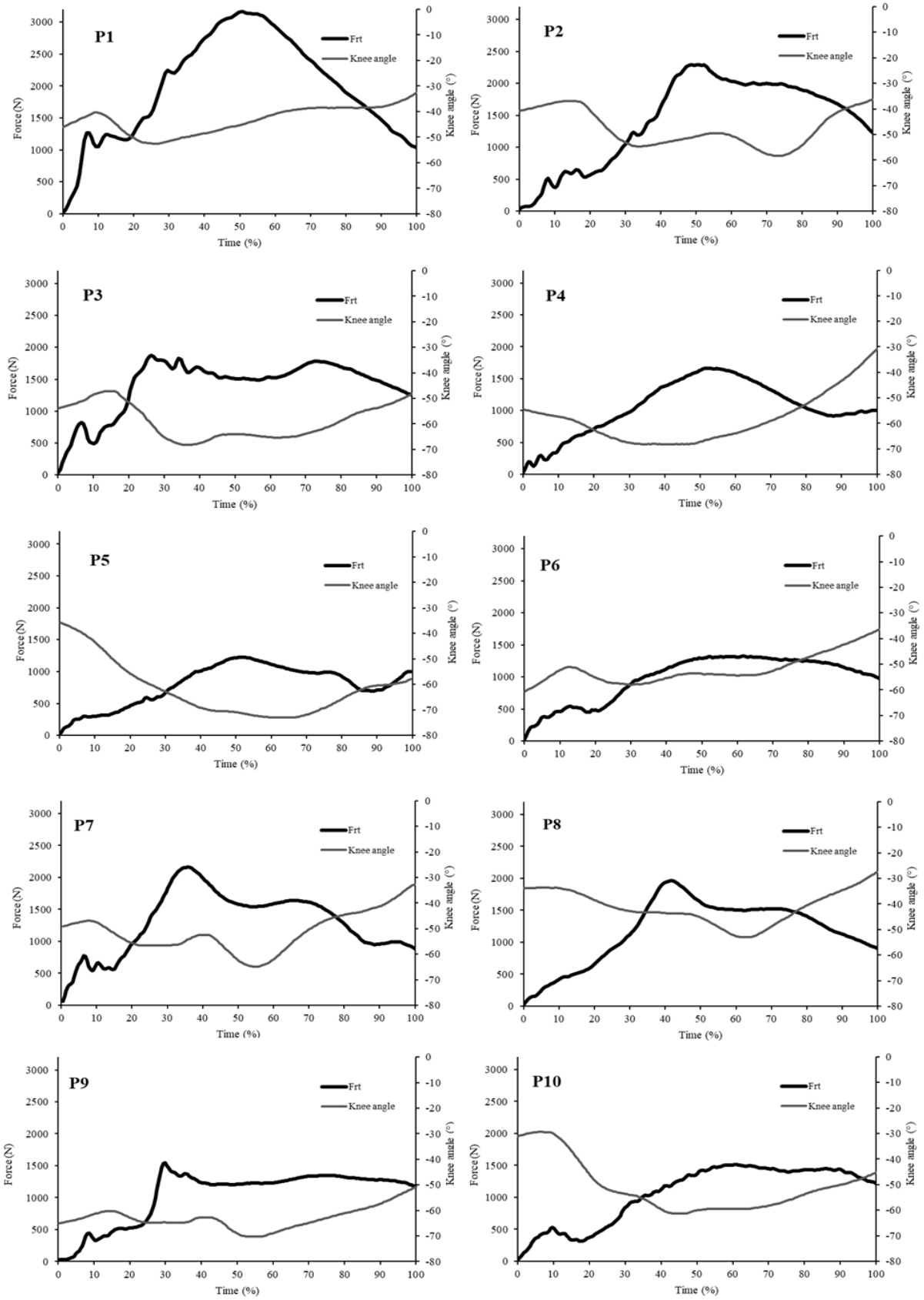
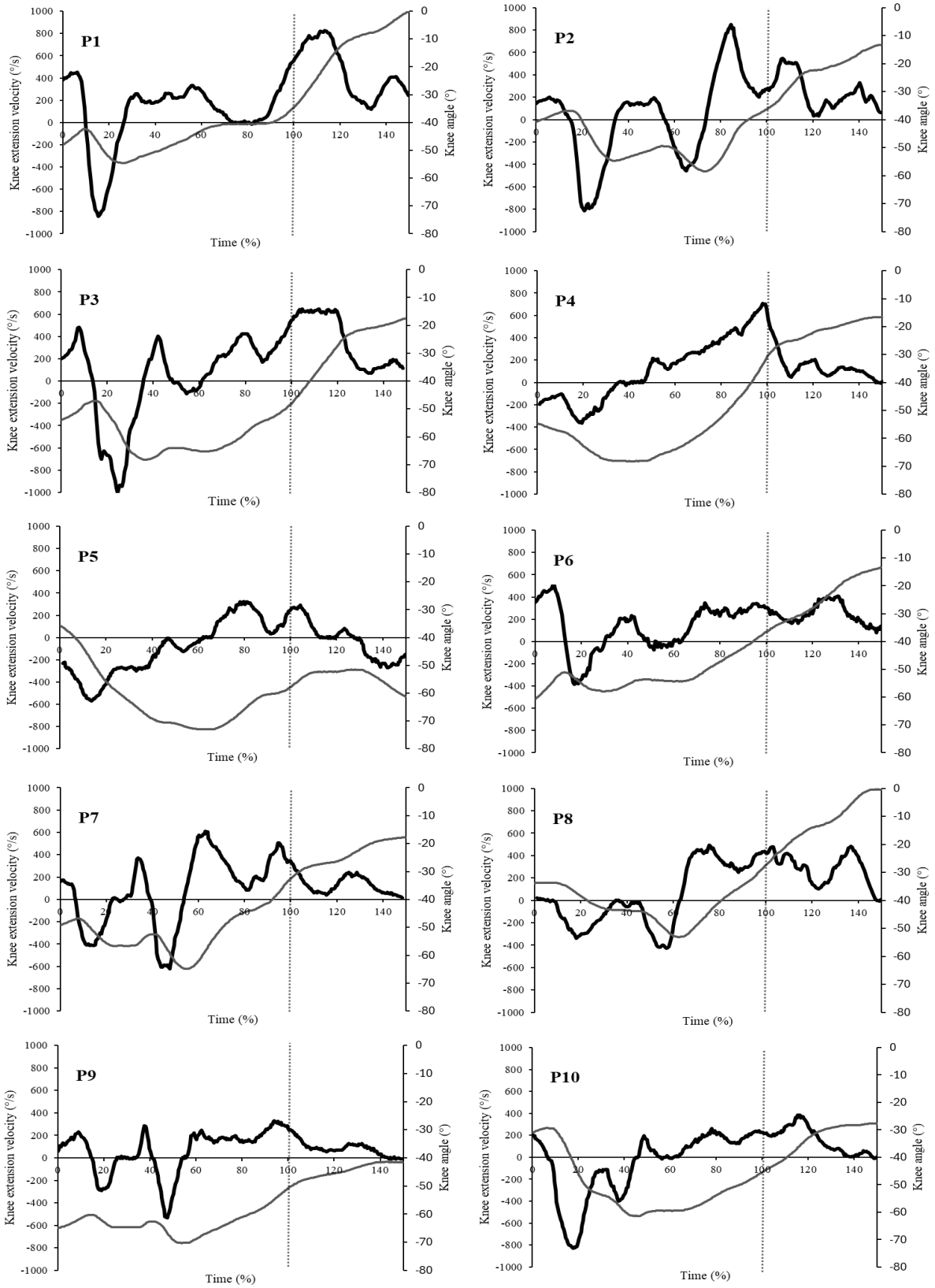
The rectangles represent the force plate. The footmarks represent the lead and rear foot positions during the last step and the lines in between illustrate their orientation relative to the force plate. A difference is seen that could be caused by a different run-up orientation or greater stance deviation. The main directions of the ground reaction forces and their orientations relative to the force plate are presented on the upper left corners. Fz is directed straight upwards, although it does not appear so in the example.



APPENDIX 2. Ground reaction force curves. Force is presented as relative (BW) values. The scale of the Y-axis is adjusted differently in each graph.



APPENDIX 3. Total resultant force (Frt) and knee angle curves. Force is presented as absolute values. The scale of the Y-axis is identical in each graph.



APPENDIX 4. Knee extension velocity and knee angle curves. The black solid line represents extension velocity, the gray solid line represents knee angle and the dashed vertical line represents disc release.